

Ahmedabad University

Bayesian Personalized Recommendation using Latent Trait Models & Expectation Propagation

CSE 516 | Group 4 | Theoretical AI
Milestone 4: Final Presentation

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Problem and Motivation

Predicting an unseen rating

- Estimate a probability for each star from 1 to 5 for a held-out user–movie pair.
- MovieLens 1M has 1,000,209 ratings across 6,040 users and 3,706 rated movies.
- Only 4.47% of possible pairs are observed. A useful model must share evidence across ratings.

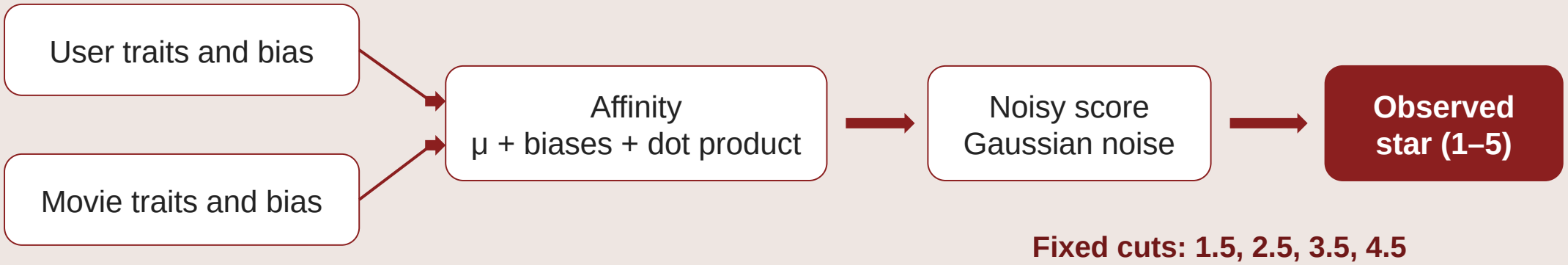
Why a probabilistic graphical model?

- Shared user and movie traits connect sparse ratings.
- An ordinal likelihood respects the order of the five stars and gives uncertainty over them.
- A movie graph can share evidence between related items with few ratings.

Research questions: How much do extra EP passes help, and what does the movie graph add?

Graphical Representation: Ordinal Latent Traits

Each user and movie has eight hidden traits and one bias. These parameters are shared across ratings.



A rating falls in one of five intervals of the noisy score.

Ratings are conditionally independent given the traits and biases. Shared parameters create dependence.

Learning: ADF and Expectation Propagation

Assumed-Density Filtering (ADF)

- Process each rating once and update the Gaussian beliefs for that user and movie.
- Keep the updated beliefs without retaining each rating’s message.
- The first EP pass equals ADF under the same order and initialization.

EP: revisiting a rating

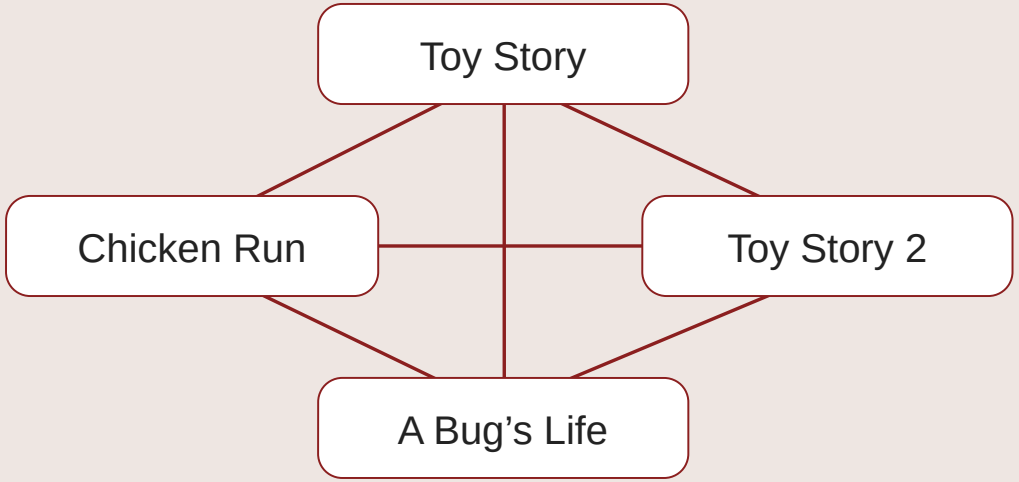
1. **Remove the old message**
   Subtract the rating’s stored contribution to get the cavity belief.
2. **Match the moments**
   Match the truncated score’s moments, then update traits with variational message passing.
3. **Replace the message**
   Damp the new contribution and add it back. EP5 and EP10 run five and ten passes.

Repeated EP keeps one site per rating, about 220 MiB here. ADF retains no sites.

Learning: Gaussian Markov Random Field

The GMRF links related movie vectors so that movies with few ratings can borrow evidence from their neighbours.

Training-only movie graph



Graph construction and MAP

- Mix residual rating similarity (90%) and genre overlap (10%). Keep the top 20 positive neighbours per movie, then symmetrize and normalize.
- The graph penalty charges for differences between neighbouring movie vectors.
- Alternate user and movie updates. Each block solve can only lower or preserve the objective.

γ controls smoothing. At γ = 0 the same MAP model and solver give the graph-free baseline.

The full graph has 3,706 nodes and about 56,508 edges (0.823% density). The diagram omits edge weights.

Inference: From a Score to Star Probabilities

ADF / EP predictions

- Combine the Gaussian trait and bias moments with observation noise.
- Approximate the affinity plus noise by a Gaussian score. Posterior covariances are omitted.
- Integrate the score over each of the five star intervals.

MAP predictions

- Use the fitted dot product and biases as the score's mean.
- Choose the Gaussian residual variance by validation log-loss.
- Use the same star intervals. Probabilities describe residual noise around the fitted score.

Use the probability-weighted mean for RMSE and MAE, and the most probable star for accuracy.

Experimental Setup, Baselines and KPIs

800,167
Train ratings per split

100,020
Validation ratings per split

100,022
Test ratings per split

Baseline comparisons

- Global, user and item means are simple rating baselines. MF-SGD is the conventional factor model.
- ADF controls for additional EP passes. Graph-free MF-MAP controls for graph smoothing.

Evaluation metrics

- RMSE and MAE: expected-rating error. Exact-star accuracy: the modal prediction.
- NLL (log-loss) and Brier score assess star probabilities. ECE compares confidence with accuracy.
- Fit time and prediction latency measure cost.

Five shared splits (seeds 42–46). Validation chooses settings. Test ratings are held out.

MovieLens 1M Test Results

Mean ± sample SD for RMSE across five splits. Other columns show means.

Method	RMSE (stars)	NLL	ECE	Exact star (%)
Item mean	0.9786 ± 0.0026	1.3209	0.0054	40.58
MF-SGD	0.8683 ± 0.0024	1.2214	0.0397	46.04
ADF / EP1	0.8977 ± 0.0035	1.2519	0.0451	44.13
EP3	0.8740 ± 0.0036	1.2283	0.0537	45.20
EP5	0.8643 ± 0.0037	1.2189	0.0566	45.59
EP10	**0.8548 ± 0.0029**	**1.2098**	**0.0608**	**46.04**
MF-MAP (no graph)	0.8506 ± 0.0021	1.2012	0.0383	46.98
GMRF-MAP	**0.8499 ± 0.0023**	**1.2007**	**0.0387**	**46.96**

EP10 reduces RMSE by 4.8% versus ADF. The lowest RMSE is 0.8499 with GMRF-MAP.

Source: M4 report, Table 1. Lower RMSE, NLL and ECE are better.

Accuracy, Calibration and Cost

Method	Fit (s)	Predict (ms / 1,000)	Brier score
MF-SGD	1.38	0.413	0.6554
ADF / EP1	3.33	0.810	0.6681
EP5	18.29	0.679	0.6552
EP10	**35.99**	**0.779**	**0.6516**
MF-MAP (no graph)	16.74	0.449	0.6467
GMRF-MAP	**14.48**	**0.487**	**0.6468**

Extra EP passes

- EP10 takes 10.8× as long to fit as ADF. Log-loss and Brier improve, but ECE rises from 0.0451 to 0.0608.

Graph workflow

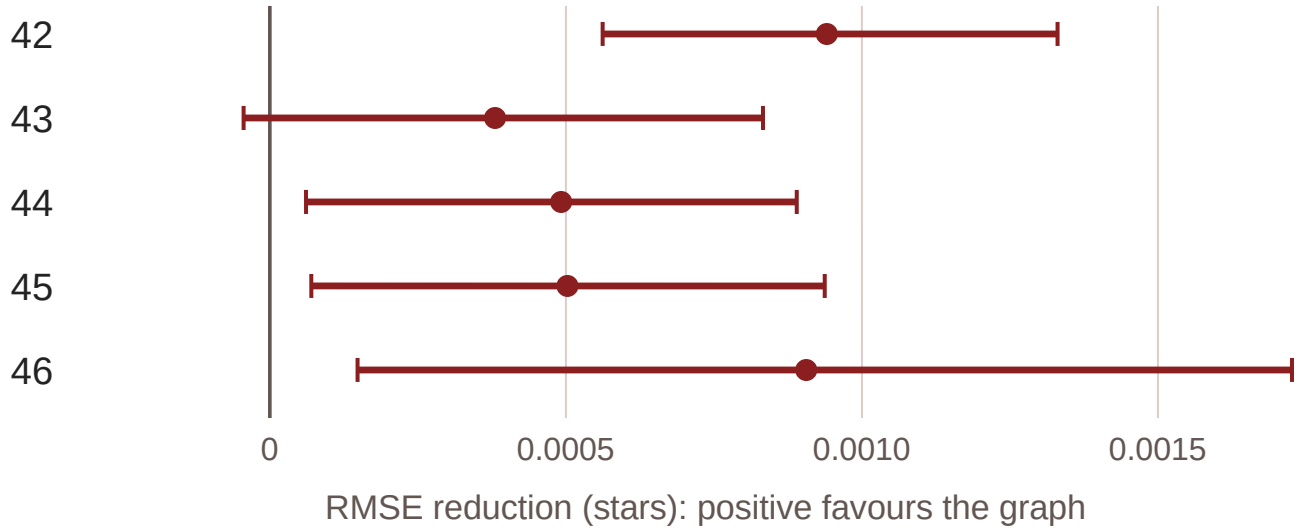
- Graph construction takes 4.31 s. The four-candidate search takes 66.37 s, including the selected 14.48 s fit.

Five-split means. Fit excludes JIT warm-up. Prediction time is per 1,000 ratings.

How Much Does the Graph Add?

The matched comparison holds the likelihood, latent dimension and MAP solver fixed.

Paired 95% intervals by split



Clean test data

- Mean gain: 0.000644 stars (0.076%).
- Validation picks γ = 10 in all five splits.
- Four intervals exclude zero; one includes it.

The graph effect is small. Intervals resample users within each split and hold the fitted models fixed.

2,000 paired resamples per split. The five splits overlap and are not independent replications.

Graph Smoothing Helps More with Weak Evidence

Mean test RMSE over three splits. The graph gain is the reduction relative to graph-free MAP.

Test condition	MF-MAP	GMRF-MAP	Graph gain
Clean training	0.8507	0.8501	0.0006
25% of training ratings kept	**0.9178**	**0.9100**	**0.0077**
20% of training stars changed	**0.9317**	**0.9248**	**0.0069**
Unseen movies	1.0622	1.0545	0.0077
Unseen users	0.9935	0.9916	0.0020

Genre links help unseen movies. For unseen users, ADF still scores better (0.9837 RMSE).

Ablation: rating-based links help on clean data. Genre-only links make clean-data RMSE worse.

Cold-start rows use different test populations; compare methods within a row. Source: M4 report, §8.1.

Algorithm Checks and Reproducibility

Reference checks

- ADF and EP1 match under the same initialization and ordering.
- In the small bilinear test, two quadrature references agree to about 10^{-10}. Hybrid inference underestimates marginal variances by about 10%.
- MAP block updates never increase the objective in the numerical checks.

Full-data replay

- A fresh seed-42 MovieLens run matched the stored split hashes.
- All 146 compared metric and selection values matched within 10^{-8} across 17 method rows, excluding timings.
- The repository contains the code, configuration, split seeds and result tables.

The small-model checks test inference mechanics. Held-out MovieLens scores test predictive performance.

Positioning Against Matchbox and PreRec

Matchbox · Microsoft Research, 2009

- Uses latent traits and hybrid EP / variational message passing for online Bayesian recommendation.
- We isolate ADF versus repeated EP in a static, ID-only model with fixed rating thresholds.
- The lower error from more passes reproduces the qualitative trade-off in a narrower setting.

PreRec · WSDM 2024

- Uses hierarchical Bayesian structure and neural encoders for cross-domain zero- and few-shot ranking.
- We test a fixed movie graph within one dataset and measure held-out rating prediction.
- The shared idea is borrowing evidence when observations are sparse.

PreRec reports Recall and revised NDCG for ranking, so its scores cannot be compared directly with our RMSE.

What We Learned and What Comes Next

What the experiments show

- Repeated EP lowers point error and log-loss versus ADF, with longer fits and worse ECE.
- Graph-free MAP already captures most of the clean-data gain. Smoothing helps more when ratings are sparse or noisy.

Where the results stop

- Random rating splits do not test prediction on future data.
- EP drops posterior covariance. MAP fixes traits and graph weights at prediction time.
- Movie links cannot personalize for users with no ratings.

Next study: use chronological splits and compare posterior-aware graph inference with a matched graph-free model.

References and Repository

[1] Stern, Herbrich and Graepel. “Matchbox: Large Scale Online Bayesian Recommendations.” WWW 2009. doi:10.1145/1526709.1526725

[2] Lin et al. “Pre-trained Recommender Systems: A Causal Debiasing Perspective.” WSDM 2024. doi:10.1145/3616855.3635779

[3] GroupLens Research. MovieLens 1M Dataset. grouplens.org/datasets/movielens/1m/

[4] Harper and Konstan. “The MovieLens Datasets: History and Context.” ACM TiiS, 2015. doi:10.1145/2827872

Project repository: github.com/Jyotier2006/M2_g4_th_apl_recommend-sys_team